



Building RAG Agents with LLMs

Search Ranking: BM25, Embeddings & Evaluation

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One Query, One Teaching Corpus

Query: solar energy storage

Document	Tokens	Length
A	solar energy storage battery	4
B	solar battery storage	3
C	energy storage overview	3
D	solar system planet	3
E	energy market prices	3
F	storage furniture box	3

N = 6 documents Average length = 19 / 6 = 3.167 tokens



Lexical Retrieval

One-hot, Bag-of-Words, TF-IDF and BM25

One-hot Encoding

One token activates one vocabulary position

Token	solar	energy	storage	battery
solar	1	0	0	0
energy	0	1	0	0
storage	0	0	1	0

One-hot represents identity. It does not learn meaning.

Bag-of-Words

Counts combine token identities into a document vector

Text	solar	energy	storage	battery
solar solar storage	2	0	1	0
storage solar solar	2	0	1	0
solar energy battery	1	1	0	1

Same counts, same vector, even when word order changes.

Term Frequency

TF counts occurrences inside one document

solar solar solar

3

TF(solar)

Repeated mention

solar battery

1

TF(solar)

One mention

Document Frequency and IDF (Inverse Document Frequency)

A common term contributes less discrimination

$$\text{IDF}(t) = \log_2 \left(\frac{N}{\text{DF}(t)} \right)$$

N

Number of documents

DF(t)

Documents containing term t

log₂

Compresses the rarity ratio

Term	DF	IDF
solar	3	$\log_2(6/3) = 1.000$
storage	4	$\log_2(6/4) = 0.585$

TF-IDF Weight

Local frequency multiplied by corpus-wide rarity

$$w_{t,d} = \text{TF}(t, d) \times \text{IDF}(t)$$

t, d

A term and a document

w(t,d)

Weight of term t in document d

TF × IDF

Frequent here, distinctive overall

Term in document A	TF	IDF	Weight
solar	1	1.000	1.000
energy	1	1.000	1.000
storage	1	0.585	0.585

Cosine Similarity

Compare vector direction after length normalization

$$\cos(q, d) = \frac{\mathbf{w}_q^T \mathbf{w}_d}{\|\mathbf{w}_q\|_2 \|\mathbf{w}_d\|_2}$$

w_q, w_d

Query and document weight vectors

Numerator

Multiply matching coordinates, then sum

Denominator

Product of full vector lengths

Step	Document A
1. Shared weighted terms	$1^2 + 1^2 + 0.585^2 = 2.342$
2. Full vector lengths	1.530×2.203
3. Divide	$2.342 / (1.530 \times 2.203) \approx 0.695$

A TF-IDF Ranking

Computed on the same six-document corpus

Rank	Document	Cosine score	Matched query terms
1	A	0.695	solar, energy, storage
2	B	0.447	solar, storage
3	C	0.310	energy, storage
4=	D / E	0.172	solar / energy
6	F	0.060	storage

Why BM25?

A lexical score with controlled term repetition

Rarity

How common is the term?

IDF weighting

Frequency

How often does it appear here?

Diminishing extra gain

Length

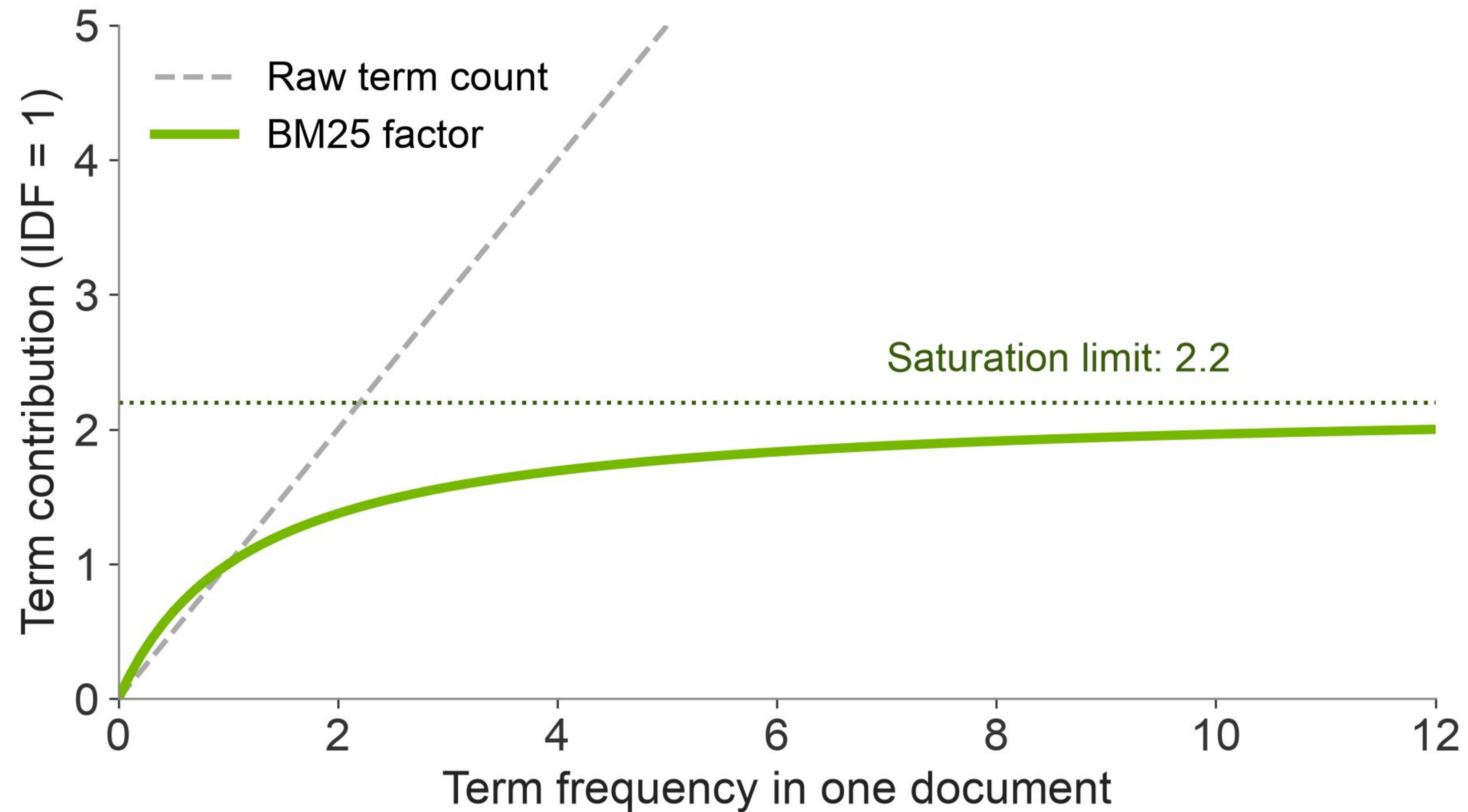
How long is this document?

Relative-length adjustment

BM25 balances rarity, repetition and document length.

BM25 Frequency Saturation

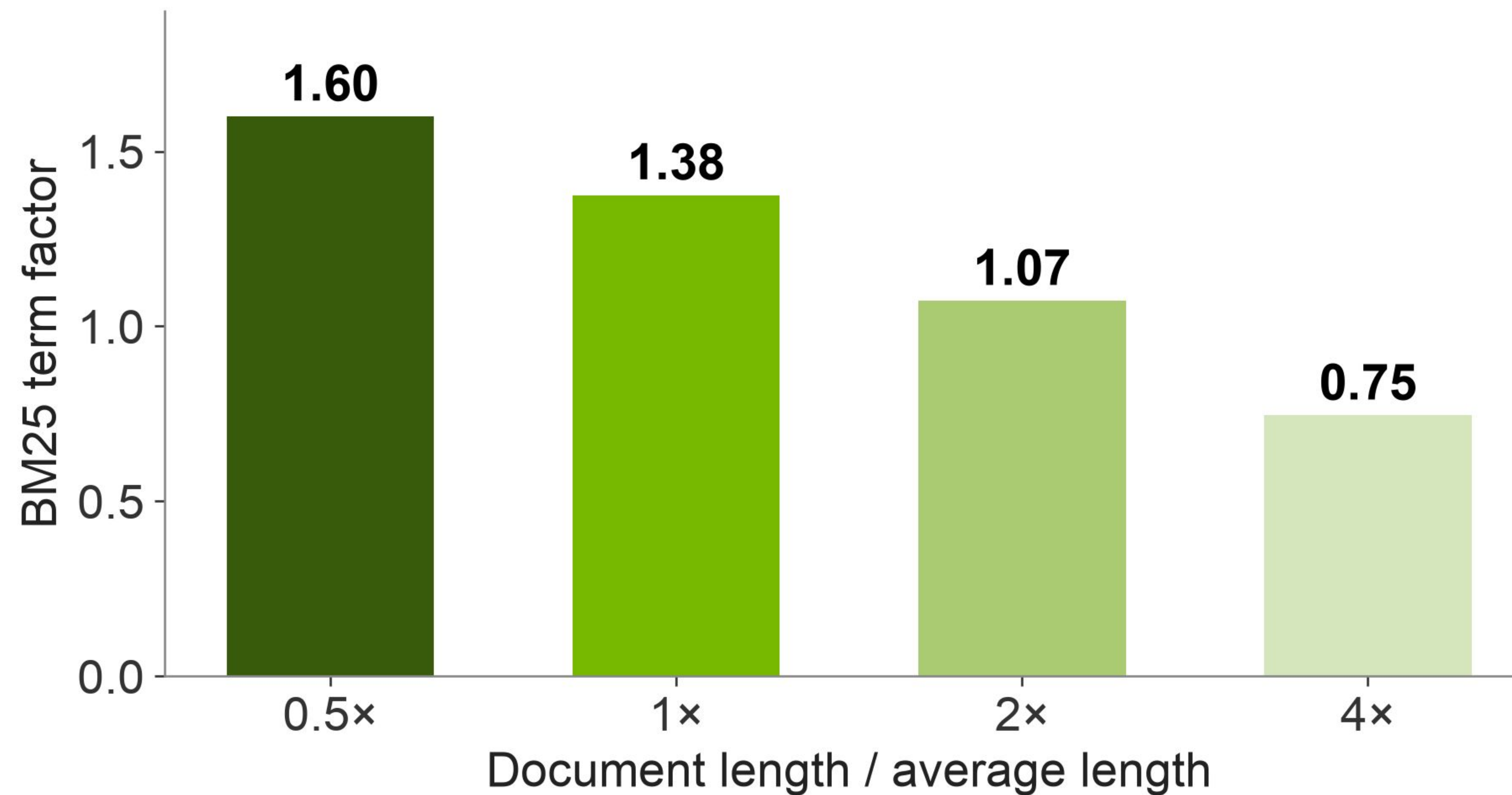
The first extra occurrence matters more than the eleventh



$k_1 = 1.2$ $IDF = 1$, length ratio = 1

BM25 Length Normalization

Two mentions carry different weight in different-length documents



b controls how strongly relative document length matters.

The BM25 Formula

One term contributes a weighted, normalized frequency

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \frac{f_{t,d}(k_1 + 1)}{f_{t,d} + k_1(1 - b + b|d|/\text{avgdl})}$$

Σ over t in q

Add the contribution of each distinct query term

IDF(t)

Rarity weight

Fraction

Frequency saturation with length normalization

Symbol	Meaning
$f(t,d), d $	Term count in d , document token count
avgdl	Average document length in the corpus
k_1, b	Frequency and length controls

Read it as: sum of rarity × adjusted frequency.



Neural Retrieval

Encoders, embeddings and candidate reranking

Static and Contextual Embeddings

A word can change meaning with its context

river bank

Land beside a river

bank account

A financial institution

Representation	Input	What changes?
One-hot	A token identity	Only its vocabulary position
Word2Vec / GloVe	A word type	Learned static vector
Contextual encoder	Tokens in a sentence	Token vectors depend on context

Contextual token vectors still need a suitable retrieval representation.

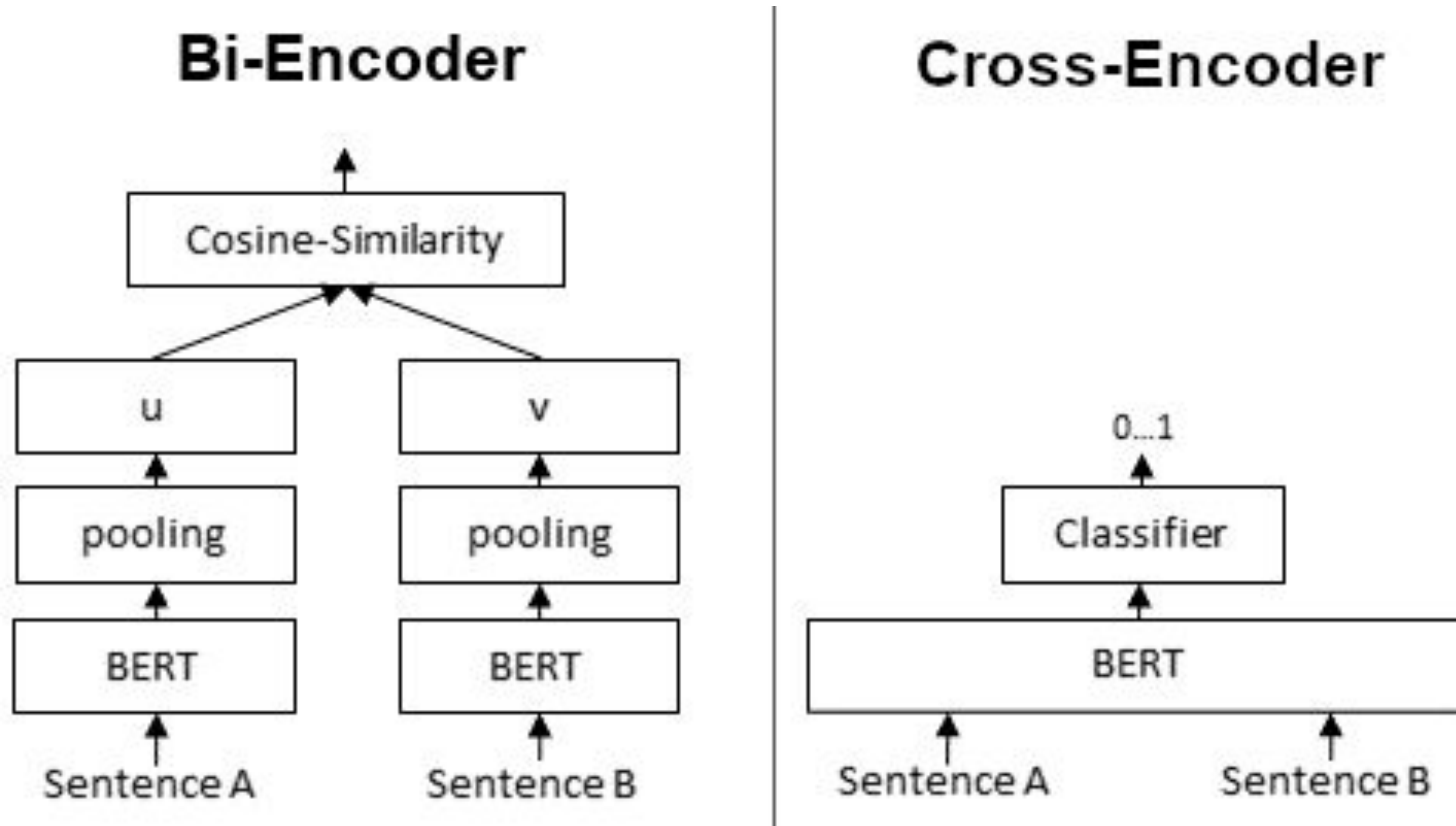
Encoder and Pooling

Token representations become a query or document embedding

- | | | |
|-----------|-----------------|---|
| 01 | Tokenize | solar / energy / storage
Convert text into model tokens |
| 02 | Encode | One hidden vector per token
Use context to represent tokens |
| 03 | Pool | Mean, CLS or last token
Produce one text vector |
| 04 | Compare | Cosine or dot product
Score query–document similarity |

Bi-Encoder vs. Cross-Encoder

Independent vectors or a joint relevance judgment



Qwen3 Embedding and Reranker

One family, two different output types

Embedding

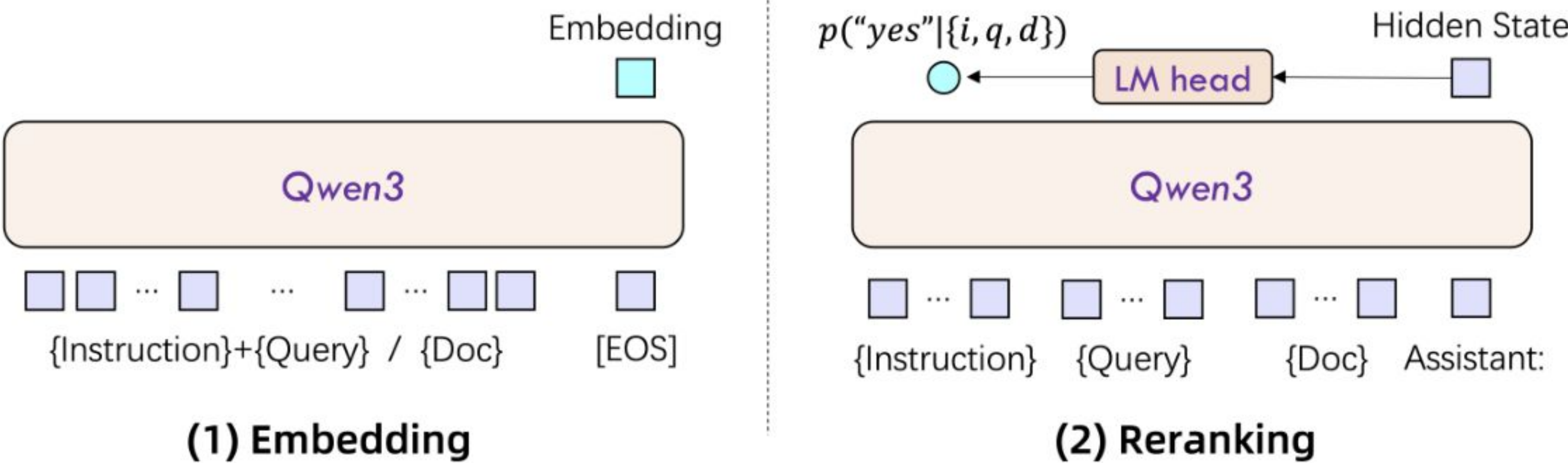
A reusable vector

Reranker

A query–document score

Sizes

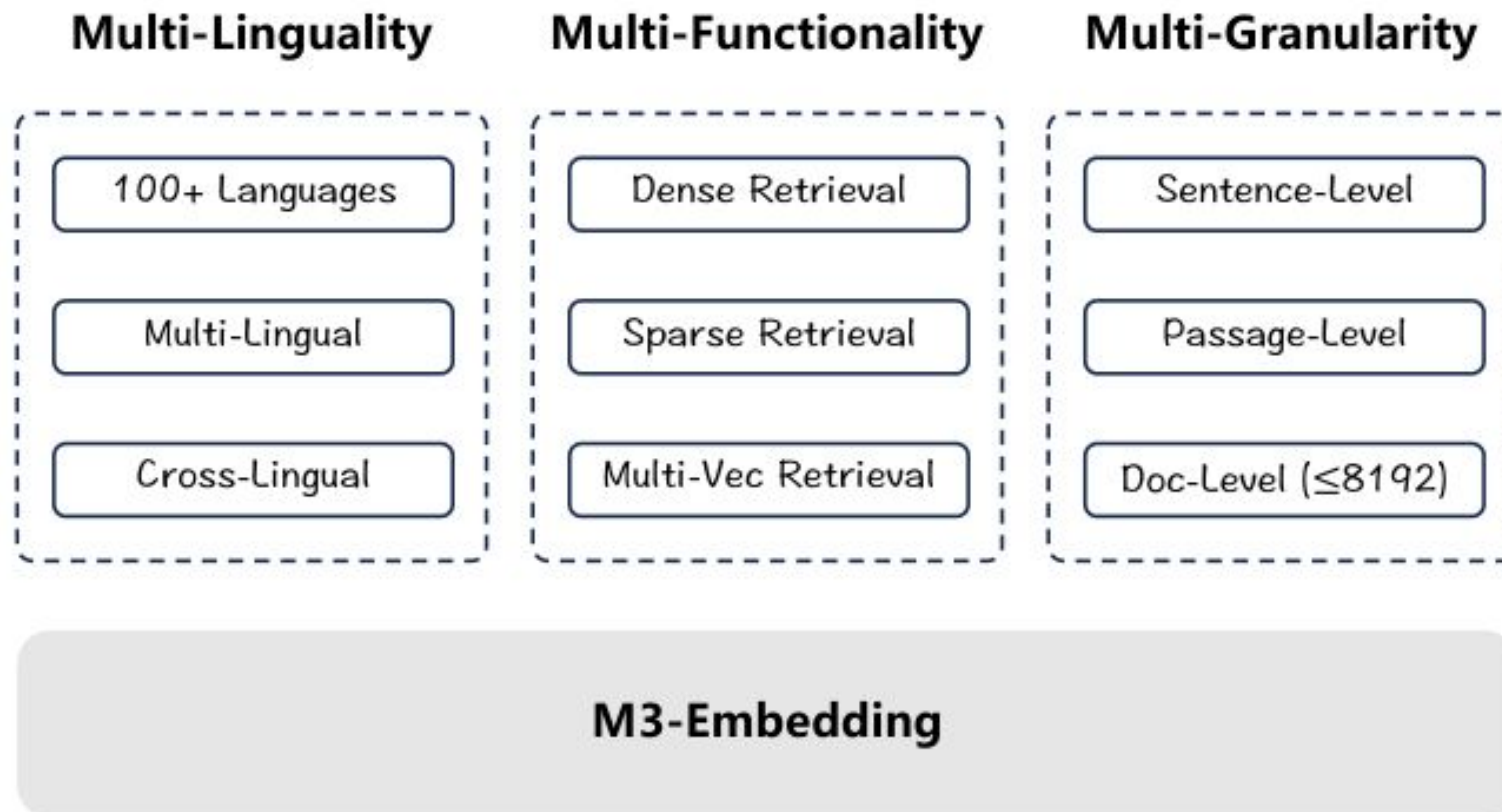
0.6B, 4B and 8B



Embedding model for indexing. Reranker model for candidate scoring.

BGE-M3

Dense, sparse and multi-vector retrieval in one model



Dense

One 1024-dimensional vector

Sparse

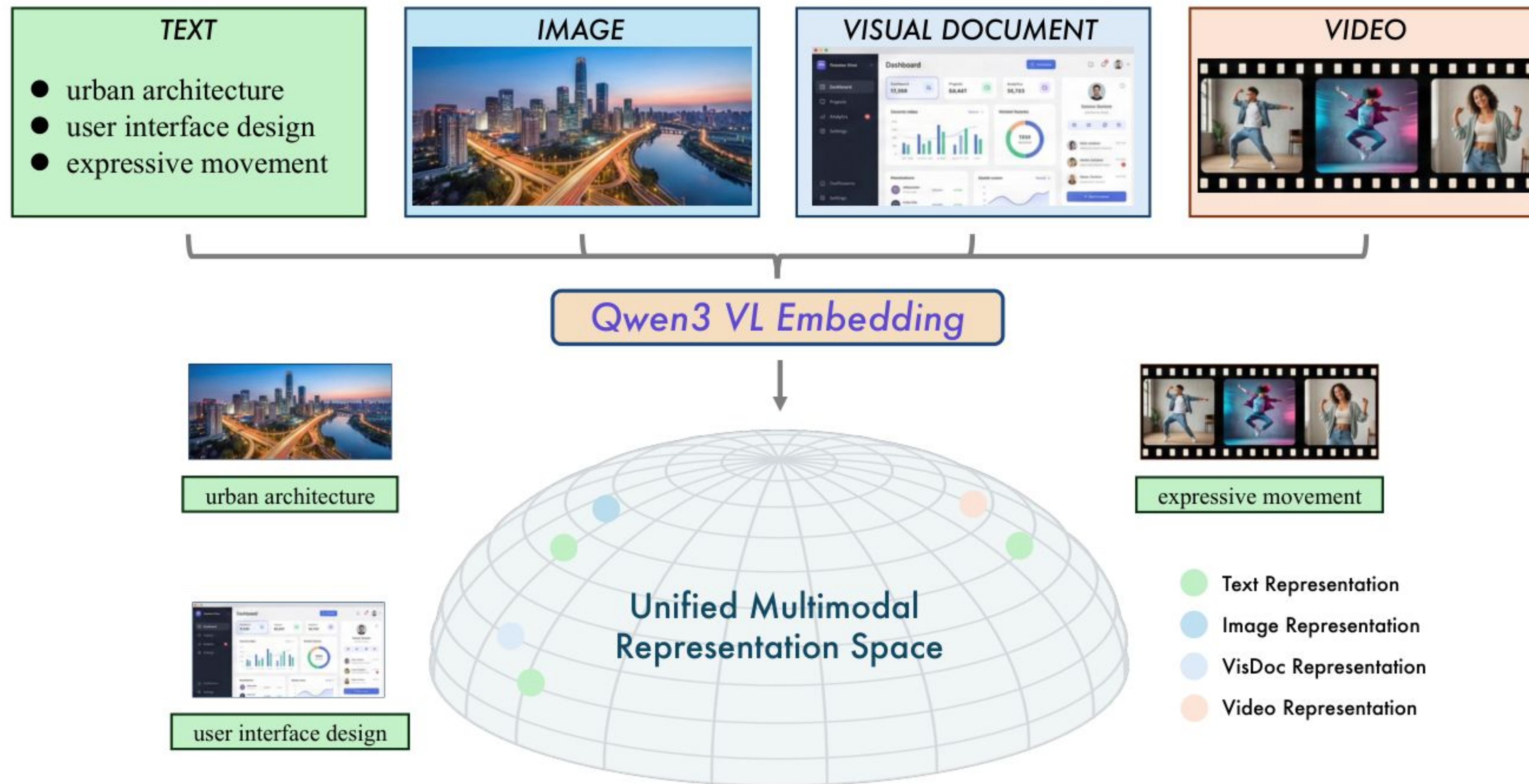
Vocabulary-based weights

Multi-vector

Token-level interaction

Multimodal Retrieval

Qwen3-VL can represent text, images, pages and video



The document's modality changes the appropriate model and index.

Cross-Encoder Scoring

Each candidate is read together with the query

Model input	Output	Reuse across different queries?
query + document A	score for A	Score must be recomputed
query + document B	score for B	Score must be recomputed

Joint scoring trades more computation for finer query–document interaction.

Reranker Families

Reranker names describe a role, not one architecture

Example	Scoring approach	Use and limitation
BGE-reranker-v2-m3	Joint query–document scoring	Multilingual candidate refinement
Qwen3-Reranker	Instruction-aware yes/no scoring	Flexible, but more compute at larger sizes
Listwise LLM reranking	Judge several candidates together	More context and ordering complexity

Match the reranker to language, input length and latency budget.

The Candidate Ceiling

A reranker can only reorder the documents it receives

Corpus: A · B · C · D · E · F

Useful document

A

RETRIEVED CANDIDATES

B · C · D

AFTER RERANKING

Still only B · C · D

A was never passed to the reranker.

Candidate recall limits downstream ranking quality.

Learning to Rank

Learn which documents should appear first

Pointwise

A document grade

A has relevance grade 3

Pairwise

Which document is better?

A should rank above B

Listwise

Quality of the whole order

A, B, C should outrank C, B, A

Training objective and evaluation metric need not be identical.



Ranking Metrics

Coverage, early success and graded usefulness

Relevance Judgments

Define useful before calculating a metric

Document	Teaching grade	Binary relevant?
A: direct help	3	1
B: useful help	2	1
C: partial help	1	1
D / E / F: not useful	0	0

The Ranking Used Below

A deliberately imperfect order for metric calculations

Rank	1	2	3	4	5	6
Document	D	A	E	B	C	F
Grade	0	3	0	2	1	0

Relevant documents appear at ranks 2, 4 and 5.

Precision@K

Among the shown results, how many are useful?

$$P@K = \frac{\sum_{i=1}^K \text{rel}_i}{K}$$

K

How many top results we inspect

rel_i

1 if rank i is relevant, otherwise 0

Numerator

Relevant results inside the top K

Rank	1	2	3	4	5
Document	D	A	E	B	C
Relevant	0	1	0	1	1

$$P@5 = 3 / 5 = 0.60$$

Strength: top-result purity. Blind spot: missed evidence and internal order.

Recall@K

Of all useful documents, how many did we retrieve?

$$\text{Recall@K} = \frac{\sum_{i=1}^K \text{rel}_i}{R}$$

R

All relevant documents in the judgment set

Numerator

The same count used in Precision@K

Denominator

All relevant documents, not all returned documents

Rank	1	2	3	4	5
Document	D	A	E	B	C
Relevant	0	1	0	1	1

$$\text{Recall@5} = 3 / 3 = 1.00$$

Strength: coverage. Blind spot: noise and ordering within top K.

Same Set, Different Reading Experience

Precision and Recall ignore order within top K

Original order

Rank	1	2	3	4	5
Document	D	A	E	B	C
Grade	0	3	0	2	1

Improved order

Rank	1	2	3	4	5
Document	A	B	C	D	E
Grade	3	2	1	0	0

Both lists: $P@5 = 0.60$ and $Recall@5 = 1.00$

Hit@K and Reciprocal Rank

One useful result, and how soon it appears

$$RR(q) = \frac{1}{r_{\text{first}}}$$

r_{first}

Position of the first relevant result

No hit

RR is 0 if no relevant result is returned

Metric	Question	Same-list example
Hit@3	Any relevant result in top 3?	Yes: 1
RR	Where is the first relevant result?	Rank 2: 1/2 = 0.50

Hit and RR focus on first success, not complete evidence coverage.

Mean Reciprocal Rank

Average the first-success score across queries

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \text{RR}(q)$$

Q, |Q|

Evaluation queries and their count

RR(q)

First-relevant-position score for one query

Mean

Each query receives equal weight

Query	First relevant rank	RR
Query 1	2	0.50
Query 2	1	1.00

$$\text{MRR} = (0.50 + 1.00) / 2 = 0.75$$

Average Precision

Read Precision only where a relevant result appears

$$AP = \frac{1}{R} \sum_{i=1}^L P@i \text{ rel}_i$$

L

Length of the evaluated result list

rel_i

Include only relevant positions

R

Total relevant documents, including any missed ones

Relevant rank	Precision at that rank	Contribution
2: document A	1 / 2	0.50
4: document B	2 / 4	0.50
5: document C	3 / 5	0.60

$$AP = (0.50 + 0.50 + 0.60) / 3 \approx 0.533$$

Mean Average Precision

First average within a query, then average across queries

$$\text{MAP} = \frac{1}{|Q|} \sum_{q \in Q} \text{AP}(q)$$

AP(q)

Average Precision for one query

|Q|

Number of evaluation queries

Equal weight

Each query contributes one AP value

Query	Relevant ranks / total relevant	AP
Query 1	2, 4, 5 / R = 3	0.533
Query 2	1, 3 / R = 2	0.833

$$\text{MAP} \approx (0.533 + 0.833) / 2 \approx 0.683$$

Binary and Graded Relevance

Some useful results are more useful than others

Grades 1, 2, 3

Rank	1	2	3
Document	C	B	A
Grade	1	2	3

Grades 3, 2, 1

Rank	1	2	3
Document	A	B	C
Grade	3	2	1

Discounted Cumulative Gain

Useful results contribute more when they appear earlier

$$\text{DCG@K} = \sum_{i=1}^K \frac{2^{g_i} - 1}{\log_2(i + 1)}$$

g_i

Relevance grade at rank i

$2^{g_i} - 1$

Gain from that grade

$\log_2(i + 1)$

Rank-based discount in the denominator

Part	Read as	Example
Gain	How useful is it?	Grade 3 gives 7
Discount	How early is it?	Rank 2 divides by $\log_2 3$
Sum	Total discounted usefulness	Add top-K contributions

DCG combines grade, position and cumulative contribution.

DCG Worked Example

The original order D, A, E, B, C

Rank	Document / grade	Gain	Discounted contribution
1	D / 0	0	0
2	A / 3	7	$7 / \log_2 3 = 4.417$
3	E / 0	0	0
4	B / 2	3	$3 / \log_2 5 = 1.292$
5	C / 1	1	$1 / \log_2 6 = 0.387$

DCG@5 = 6.095 using full precision before rounding.

Normalized DCG

Compare actual gain with the best possible ordering

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$

DCG

Discounted gain of the actual ranking

IDCG

DCG of the ideal order for the same judgments

Ratio

Share of ideal discounted gain

Ordering	Grades	DCG@5
Actual: D A E B C	0, 3, 0, 2, 1	6.095
Ideal: A B C D E	3, 2, 1, 0, 0	9.393

$$\text{NDCG@5} = 6.095 / 9.393 \approx 0.649$$

What Changes After Reranking?

Same top-5 set, improved positions

Metric	D A E B C	A B C D E	What it notices
P@5	0.60	0.60	Useful count
Recall@5	1.00	1.00	Coverage
RR	0.50	1.00	First useful rank
AP	0.533	1.000	All relevant positions
NDCG@5	0.649	1.000	Grades and positions

The metric should match the improvement you want to measure.

Metric Strengths and Blind Spots

A score answers one particular question

Metric	Useful for	Does not fully describe
Precision@K	Top-result purity	Missed documents and order
Recall@K	Evidence coverage	Noise and order
Hit@K / MRR	First useful answer	All evidence and graded utility
MAP	Ordering multiple relevant results	Graded usefulness
NDCG@K	Graded usefulness near the top	Latency, cost or factual answer quality

Metric selection starts with the intended search experience.



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